

REPORT

FOR

The World Packaging Organisation (WPO)



Better quality of life, through
better packaging, for more people

BY

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Executive Summary

This report presents a high-level assessment of the packaging value chain in Uganda. It examines how packaging materials are sourced, manufactured, distributed, used, regulated, and managed after use. The assessment identifies key actors, structural gaps, trade patterns, sustainability challenges, and opportunities for sector growth.

Packaging plays an important role in Uganda's economy. It protects food, reduces post-harvest losses, supports agro-processing, enables exports, and improves product quality. It is especially important for micro, small, and medium enterprises (MSMEs), which make up most businesses in the country. Uganda has active packaging conversion capacity, particularly in plastics and paper-based products. However, the country depends heavily on imported raw materials such as plastic resins, kraft paper, tinplate, and glass containers. This creates trade imbalances and exposes the sector to exchange rate risks. There is no stand-alone Packaging Act. Packaging is guided through industrial, trade, environmental, and MSME policies. Standards are regulated mainly by the Uganda National Bureau of Standards (UNBS), but detailed technical guidance for advanced packaging remains limited. Waste management and recycling systems are still developing. A large share of plastic waste is not formally recovered, and the informal sector plays a major role in recycling. Infrastructure gaps and weak enforcement limit progress toward a circular economy. Despite these challenges, Uganda has strong potential. The country has natural resources for glass production, agricultural residues for biodegradable materials, and growing regional markets under the East African Community and the African Continental Free Trade Area. With targeted policy reform, investment in upstream industries, stronger recycling systems, and improved technical capacity, Uganda can strengthen its packaging sector and improve competitiveness, sustainability, and export performance. This report provides an overview of the packaging value chain in Uganda, focusing on the mapping and capacity assessment of the sector. The objective of this study is to identify the key players, industries, and challenges in the packaging value chain, with a view to informing policy and investment decisions that can enhance the competitiveness and sustainability of the sector.

1. Introduction

Packaging is a critical component of the industry in Uganda, playing a significant role in ensuring product safety, enhancing export competitiveness, and promoting branding and sustainability. The Ugandan government, through various initiatives, has recognized the importance of packaging in driving economic growth and sustainable development. Uganda, under Ministry of Trade Industries and Cooperatives established the IPPU to contribute to the packaging sector development by bringing together of the packaging sector actors together to forge the ways for the sector.

Uganda's packaging sector continues to grow, there is a pressing need for innovative and sustainable packaging solutions that reduce negative environmental impact, improve the quality and safety of locally packaged goods, and enhance market access domestically, regionally and internationally. This Packaging Value Chain Assessment examines the structure and performance of Uganda's packaging sector and how it supports the country's broader industrial and economic development goals. The assessment looks at the full packaging ecosystem, including the sourcing of raw materials, conversion and manufacturing processes, distribution systems, product use, regulatory oversight, and the management of packaging waste.

Packaging is an essential but often overlooked component of economic development. Effective packaging protects products from contamination and physical damage, extends shelf life, improves transport efficiency, and helps maintain product quality throughout the supply chain. In Uganda, where agriculture and agro-processing are major drivers of the economy, packaging plays a particularly important role in reducing post-harvest losses, improving product presentation, and enabling businesses to access formal retail and export markets.

For farmers, processors, and small businesses, packaging can significantly increase product value and market competitiveness. Proper packaging also ensures that products meet food safety, labeling, and quality requirements required by both domestic regulators and international markets.

At the global level, the packaging sector is undergoing significant transformation. Growing concerns about plastic pollution, climate change, and resource efficiency are pushing governments and industries to rethink how packaging is produced, used, and managed. Many countries are transitioning toward circular economy approaches that prioritize waste reduction, recycling, and the responsible management of materials through systems such as Extended Producer Responsibility (EPR).

Uganda is beginning to move in this direction, but several structural constraints remain. These include high reliance on imported packaging materials, limited local production of upstream inputs, evolving regulatory frameworks, and gaps in waste management infrastructure.

Against this backdrop, understanding the current state of the packaging value chain is essential. A clear and evidence-based assessment can help identify opportunities for investment, policy reform, and technical support that will strengthen the sector and enable it to better contribute to industrial growth, trade competitiveness, and environmental sustainability.

1.1 Objectives

The overall objective of this assessment is to provide a comprehensive understanding of the structure and performance of Uganda's packaging value chain, and to identify areas where targeted interventions can support sector development. The proposed activity aims to map the packaging value chain in Uganda and assess the capacity of the sector to identify areas for improvement and opportunities for investment.

Specifically, the assessment aims to:

- i. Map the structure of Uganda's packaging value chain
- ii. Identify and categorize the main actors involved across the sector including suppliers, converters, MSMEs, regulators, testing bodies, waste management stakeholders and academia.
- iii. Review the regulatory and policy environment governing packaging, including standards, environmental regulations, and industrial development frameworks
- iv. Assess the current state of sustainability and waste management systems, including recycling activities and the role of the informal sector.
- v. Identify key gaps, constraints, and strategic opportunities that could support the growth, competitiveness, and sustainability of Uganda's packaging sector.

2.0 Methodology

A mixed-methods approach was employed to gather data and information on the packaging value chain in Uganda. This included a combination of desk research, key informant interviews, and surveys. A review of existing literature on the packaging industry in Uganda was conducted to gather information on the sector's history, growth, and challenges. Key informant interviews were conducted with stakeholders in the packaging value chain. A survey was conducted among others packaging manufacturers, suppliers, academia, testing bodies, regulators etc., to gather information on their capacity, production levels, and challenges. A total of 25 manufacturers were reached.

3.0 SECTOR OVERVIEW

Uganda's packaging industry stems from traditional and local materials, such as sisal, clay, and handmade paper, which served small-scale industries and local markets. The sector came into drive in the early 2000s with the mushrooming of agro-processing, export-oriented industries, and increased foreign investment. The development of packaging standards and regulations by the Uganda National Bureau of Standards (UNBS), has helped to formalize the sector. In the past two decades, the industry has transitioned towards modern packaging solutions in line with international quality and safety standards, driven by export demands and rising domestic consumption. The sector is experiencing a shift towards sustainable and innovative packaging solutions. Flexible packaging (such as pouches and wraps) is gaining popularity due to its cost, versatility and convenience. The demand for eco-friendly biodegradable plastics and recycled paper, is increasingly driven by consumer awareness and international environmental concerns. New technologies such as digital printing technology are being adopted for branding and customization, especially in food and beverage packaging sector. The rise of e-commerce and retail sectors has further fueled demand for innovative packaging solutions that enhance product shelf life and delivery efficiency. Innovation efforts are supported by both government initiatives and international development programs like UNIDO. Local companies are investing in research to develop biodegradable plastics and recycled packaging materials. Adoption of digital printing techniques allows for better branding and marketing. There is also a growing interest in smart packaging, including tamper-evident and QR code-enabled packaging, to improve traceability and consumer engagement. These innovations aim to reduce environmental impact while increasing competitiveness in regional and international markets. The packaging sector directly employs approximately 50,000 people, including factory workers, designers, quality controllers, and distribution

personnel. Indirect employment includes supply chain actors like raw material suppliers, logistics providers, and recyclers. The sector contributes about 2.5% to Uganda's GDP, estimated at USD 600 million annually, according to the Uganda Bureau of Statistics (UBOS). As modernization and innovation continue, the sector's contribution is expected to grow, especially with increased export activities and local value addition. Uganda imports of packaging materials mainly consisting of plastics (about 60%), cardboard, metal, and glass packaging. The main sources of imports are China, India, Kenya, and South Africa. Uganda relies heavily on imported packaging materials and inputs. Trade data indicates that in 2024, imports of plastics and articles thereof exceeded USD 635 million, while paper and paperboard imports reached approximately USD 175 million. During the same period, plastic exports were valued at about USD 62 million, highlighting a significant trade imbalance and the country's continued dependence on imported materials. The government collects import duties estimated at USD 20 million annually, contributing to revenue and trade balance. These figures reflect a growing dependence on imported packaging, highlighting opportunities for local manufacturing expansion to reduce reliance on imports and improve trade sustainability. The reliance on imported inputs exposes the sector to fluctuations in global commodity prices and foreign exchange rates, which can affect production costs and the competitiveness of local packaging converters. At the same time, the gap between domestic demand and local production capacity highlights potential opportunities for investment in upstream packaging industries. Uganda's packaging industry is being aligned with global sustainability trends. Policies promote the use of biodegradable plastics, recycling, and waste reduction to address plastic pollution. The government, through the National Environment Management Authority (NEMA), encourages industries to adopt circular economy principles. Local initiatives include recycling programs and awareness campaigns. The industry's shift towards eco-friendly materials positions Uganda as a responsible actor in the global effort to reduce plastic waste, aligning with international commitments such as the Sustainable Development Goals (SDGs). Uganda's packaging sector is active but remains fragmented. The country has developed growing conversion capacity, particularly in plastics, flexible packaging, and paper-based cartons. However, upstream industries such as petrochemicals, pulp and paper processing, and glass container manufacturing are still limited, which constrains the development of a fully integrated packaging value chain. Demand for packaging in Uganda is largely driven by micro, small, and medium enterprises (MSMEs), agro-processors, food and beverage companies, cosmetics manufacturers, pharmaceutical firms, and producers of chemicals and agricultural inputs. As these sectors expand, the demand for reliable, safe, and market-ready packaging solutions continues to grow.

3.1 Policy and Regulatory Strategic Framework for Packaging

Uganda's packaging sector operates within a broad policy and regulatory environment that supports industrial development, export competitiveness, MSME growth, and environmental sustainability. Packaging is increasingly recognized as an important component of value addition, product safety, and trade facilitation, enabling businesses to meet market standards and compete both locally and internationally.

While Uganda does not yet have a standalone Packaging Act, the sector is guided through several national policies and strategic frameworks related to industrialization, trade, environmental management, and standards. These frameworks also align with regional and continental trade systems, including the East African Community (EAC) and the African Continental Free Trade Area (AfCFTA), which influence packaging requirements for cross-border trade and product competitiveness.

3.2 Uganda Vision 2040 and Industrial Transformation

Uganda Vision 2040 outlines the country's long-term strategy to transform Uganda from a predominantly agrarian and low-income economy into a competitive upper-middle-income

country. Central to this vision is the expansion of industrial production, increased value addition to agricultural products, and stronger participation in regional and global trade. Within this broader development agenda, packaging plays an enabling though often under-recognized role in supporting industrial transformation. As Uganda continues to prioritize agro-industrialization and manufacturing growth, packaging becomes a critical link between production and markets. Effective packaging protects products, extends shelf life, and improves logistics efficiency across supply chains. This is particularly important for agro-processing industries, which rely on packaging to maintain product quality, reduce post-harvest losses, and meet the regulatory and labeling requirements of domestic and export markets.

The Vision 2040 also emphasizes the importance of developing competitive manufacturing industries and reducing structural constraints that limit private sector growth. Uganda's manufacturing sector has historically been constrained by limited technological capacity, dependence on imported inputs, and relatively low levels of industrial diversification. In this context, strengthening the domestic packaging industry can contribute to industrial development in several strategic ways.

First, a well-developed packaging sector supports value addition across multiple industries including food processing, pharmaceuticals, cosmetics, and consumer goods. Packaging enables producers to differentiate their products, improve branding, and access higher-value markets. Second, it supports export competitiveness, particularly as international markets increasingly require compliance with strict packaging, labeling, and sustainability standards. Third, local packaging production can reduce the country's heavy reliance on imported packaging materials, which currently exposes businesses to foreign exchange fluctuations and global supply disruptions.

Vision 2040 also highlights the need to expand industrial infrastructure and strengthen linkages between agriculture and manufacturing. Packaging sits directly at this intersection. A stronger domestic packaging ecosystem supported by investments in raw material processing, conversion technologies, and recycling systems can stimulate upstream and downstream industrial activities while creating employment opportunities.

Finally, the Vision places growing emphasis on sustainable resource use and environmental management. As global markets shift toward environmentally responsible packaging solutions, Uganda has an opportunity to develop innovative materials using locally available resources, including agricultural residues such as banana fiber, rice husks, and sugarcane bagasse. Developing such solutions could position Uganda to participate in emerging markets for sustainable and biodegradable packaging.

3.3 National Development Plan III & IV (NDP III & IV)

Uganda's long-term development agenda, guided by Uganda Vision 2040 and implemented through the National Development Plans III & IV, aims to transform the country from a primarily agricultural, low-income economy into a competitive upper middle-income nation by 2040. Industrial growth, value addition, and export expansion are central pillars, with the industrial sector targeted to contribute over 30% of GDP by 2040.

Packaging plays a strategic, though often under-recognized, role in this transformation. For example:

- Uganda imports over USD 635 million in plastics and plastic packaging and USD 175 million in paper and paperboard annually, while exports remain limited (plastic exports ~USD 62 million), highlighting strong domestic demand but limited upstream production.
- MSMEs, which constitute over 90% of private sector businesses rely on packaging for value addition, shelf-life extension, food safety, and compliance with export standards.

- Proper packaging can reduce post-harvest losses by up to 30%, directly supporting food security and industrial growth.

Investing in local packaging capacity strengthens agro-processing and manufacturing value chains, enhances branding and product differentiation, and reduces dependence on imported materials. By aligning packaging development with Uganda's national strategies, the country can improve standards compliance, promote sustainable production, and increase competitiveness in regional and global markets.

3.4 STANDARDS ENVIRONMENT

Uganda's packaging sector operates within a complex regulatory and standards framework aimed at ensuring product safety, quality, environmental protection, and trade compliance. While there is no standalone Packaging Act, packaging activities are regulated through a combination of industrial, trade, environmental, and MSME policies.

3.4.1 Key Regulatory Bodies

- Uganda National Bureau of Standards (UNBS)**; The principal standards authority. This body is responsible for product certification (Q & S-Mark), labeling compliance, packaging standards and inspections. UNBS also provides limited guidance on export-oriented packaging requirements.
- National Environment Management Authority (NEMA)**; Regulates environmental compliance, including Extended Producer Responsibility (EPR), waste management, and environmental impact of packaging materials.
- Ministry of Trade, Industry, and Cooperatives (MTIC)**; Oversees industrial policy, trade facilitation, and competitiveness, indirectly influencing packaging requirements especially for exports.
- Local Governments & Kampala Capital City Authority (KCCA)**; Manage solid waste collection, recycling, and local environmental enforcement.

3.4.2 Standards Relevant to Packaging

Packaging in Uganda is subject to a combination of general product standards, material standards, and specific packaging standards including

- UNBS Standards for Packaging Materials**
 - US EAS 72:2010 – Corrugated fiberboard cartons
 - US EAS 48:2000 – Flexible plastic packaging
 - US EAS 46:2000 – Glass containers for food products
 - US EAS 27:2010 – Paperboard and paper-based packaging
- Labeling & Food Safety Standards**
 - UNBS QMS 220:2010 – Food product labeling
 - Codex Alimentarius standards
- Environmental Standards**
 - Extended Producer Responsibility for plastic waste management
 - Solid Waste Management Standards – For local governments and informal recycling sector

Uganda's packaging sector faces challenges including regulatory fragmentation, where multiple agencies (UNBS, NEMA, MTIC and local authorities) issue overlapping rules that create confusion for MSMEs, limited technical guidance on materials, digital printing, and export-oriented compliance; low awareness of international requirements such as the EU Packaging and Packaging Waste Regulation (PPWR), which constrains market access; and insufficient in-country testing capacity, with only a few specialized labs like UNBS. Addressing these gaps through clearer regulations, enhanced technical support, and expanded testing facilities would improve MSME compliance, strengthen export competitiveness, and support sustainable packaging practices.

4.0 POLICY AND REGULATORY GAPS

The packaging sector is governed various policies and regulations designed to regulate standards, promote environmental sustainability, and facilitate trade. These policies are administered by different government agencies, each contributing to defined aspects of the sector. The main actors in regulatory frameworks is the Uganda National Standards (UNBS Standards), managed by the Uganda National Bureau of Standards (UNBS) under Ministry of Trade Industries and Cooperatives. These standards set the requirements for packaging materials, labeling, and certification processes. Environmental policies that are particularly related to waste management and plastic pollution, are overseen by the Ministry of Environment and the National Environment Management Authority (NEMA). Regulations such as the Extended Producer Responsibility (EPR) and policies addressing plastic bans aim to promote eco-friendly packaging solutions are in place. Trade and export policies, coordinated by the Uganda Export Promotion Board (UEPB) and the Ministry of Trade, influence packaging through export standards and documentation requirements. The Uganda's industrial policies, managed by the Ministry of Trade and the Ministry of Industry, support manufacturing growth and MSME development. Skills development policies, overseen by the Ministry of Education and the Ministry of Trade, are vital for building technical capacity within the packaging sector.

4.1 Policy gaps

Although Uganda has several policies that influence the packaging sector, important gaps remain that limit the sector's growth and effectiveness. One key gap is the lack of a dedicated national packaging strategy to guide industry development, innovation, and investment. Packaging is currently addressed indirectly through broader industrial, trade, environmental, and MSME policies, which means sector-specific priorities such as technology upgrading, packaging research, and skills development are not clearly articulated. Another challenge is the limited coordination between regulatory institutions. Different agencies oversee standards, environmental compliance, and industrial development, but coordination mechanisms are not always clear. This can create compliance burdens for businesses, particularly MSMEs that may lack the technical capacity to navigate multiple regulatory requirements.

There is also a policy gap in supporting packaging innovation and technology adoption. Many local manufacturers and converters face difficulties accessing modern machinery, advanced printing technologies, and specialized packaging materials. Without targeted incentives or industrial support programs, upgrading production capacity remains slow.

Table 1 : Policies affecting the packaging sector in Uganda

Policy	Relevant Agencies	Areas of the Policy Affecting Packaging Sector	Key Impacts on Packaging Sector
Uganda National Environment Policy (2017)	Ministry of Water and Environment, National Environment Management Authority (NEMA)	Environmental management and waste reduction, pollution control, recycling standards	Promotion of eco-friendly packaging, waste management regulations, reduction of plastic pollution

Policy	Relevant Agencies	Areas of the Policy Affecting Packaging Sector	Key Impacts on Packaging Sector
Uganda Standards and Quality Policy	Uganda National Bureau of Standards (UNBS)	Standardization, quality assurance, product conformity, packaging standards	Enforcement of packaging material standards, quality control, certification processes
National Industrial Policy (2015)	Ministry of Trade, Industry and Cooperatives	Industrial growth, manufacturing development, import substitution	Promotion of local packaging manufacturing, import regulations for packaging materials
Plastic Bag Ban Policy (2019)	National Environment Management Authority (NEMA), Ministry of Environment	Banning of plastic bags, promoting eco-friendly alternatives	Reduction of plastic bag usage, encouragement of biodegradable packaging options
Waste Management Policy (2014)	Ministry of Water and Environment, NEMA	Waste collection, recycling, disposal standards	Incentivizing recycling, proper disposal of packaging waste
National Climate Change Policy (2021)	Ministry of Environment	Climate resilience, sustainable resource use, reduction of carbon footprint	Encouragement of sustainable packaging materials, reduction of carbon emissions
Consumer Protection Law (2010)	Uganda National Bureau of Standards (UNBS), Uganda Consumer Protection Authority	Consumer safety, labelling, packaging disclosures	Regulations on packaging labels, safety standards, and consumer rights protection

Policy	Relevant Agencies	Areas of the Policy Affecting Packaging Sector	Key Impacts on Packaging Sector
Buy Uganda Build Uganda (BUBU) Policy	Ministry of Trade, Industry and Cooperatives	Strengthen local content and procurement	Incentivizes government and private sector to source locally, increasing demand for domestic packaging solutions and raw materials.
MSME Policy	Ministry of Trade, Industry and Cooperatives	Support micro, small, and medium enterprises development	Promotes entrepreneurship, leading to the emergence of innovative packaging startups that can introduce eco-friendly and cost-effective solutions..

Sustainability is increasingly recognized as a priority, policies supporting circular economy practices in packaging but are still developing. Recycling infrastructure, incentives for sustainable materials, and clear implementation frameworks for waste management systems remain limited. Strengthening these areas would help improve environmental performance while supporting a more resilient and competitive packaging sector. Additionally, enforcement of packaging standards remains inconsistent, and there is a wide lack of awareness among MSMEs about certification procedures. This gap limits compliance and hampers the sector's ability to meet international standards. There is a gap in the implementation of policies, combined with the lack of clear guidelines for compliance. As the policies aim to facilitate access to international markets, MSMEs often lack the capacity and awareness to fully comply with export packaging standards, hence limiting their competitiveness. Interns of capacity building, there is a deficiency of sector-specific technical training programs, which hinder the development of skilled personnel for designing, testing, and implementing sustainable packaging solutions. Generally these policies tend to lack specific focus on the packaging sector, resulting in limited targeted incentives, technical support, and innovation promotion for packaging manufacturers.

4.2 PACKAGING AND EXPORT COMPETITIVENESS

Packaging plays an important role in determining how products compete in both local and international markets. Beyond protecting goods, packaging influences product visibility, consumer perception, branding, and compliance with regulatory requirements. For many Ugandan MSMEs, packaging has improved in recent years, particularly in the food, coffee, and cosmetics sectors. However, a gap still exists between basic functional packaging and the high-quality retail-ready packaging required in modern supermarkets and export markets.

Packaging competitiveness is also influenced by finishing techniques that improve appearance and durability. These include lamination, matte or gloss finishes, embossing, foil stamping, and high-resolution printing. Such finishes enhance brand presentation and shelf appeal but access to these technologies remains limited for many MSMEs due to cost and technical capacity constraints. Inclusivity is becoming an important aspect of modern packaging design. Features such as clear typography, strong color contrast, and Braille labeling can help visually impaired consumers identify products independently. While inclusive packaging is still limited in Uganda, adopting such practices can improve accessibility, strengthen brand responsibility, and align local products with emerging global standards.

5.0 MAPPING OF PACKAGING VALUE CHAIN IN UGANDA

A stakeholder mapping exercise was conducted to identify key actors involved in Uganda's packaging value chain. The mapping covered companies involved in packaging production, raw material supply, conversion, labeling and printing, recycling, laboratory testing, regulation, and industry representation. In total, over 25 organizations were identified across the value chain.

The results show that Uganda has a relatively active packaging conversion sector, particularly in plastics and paper-based packaging. However, the sector is dominated by converters and MSMEs, with limited upstream raw material production and relatively few specialized testing facilities.

Table 2: Mapping of Key Actors in Uganda's Packaging Value Chain

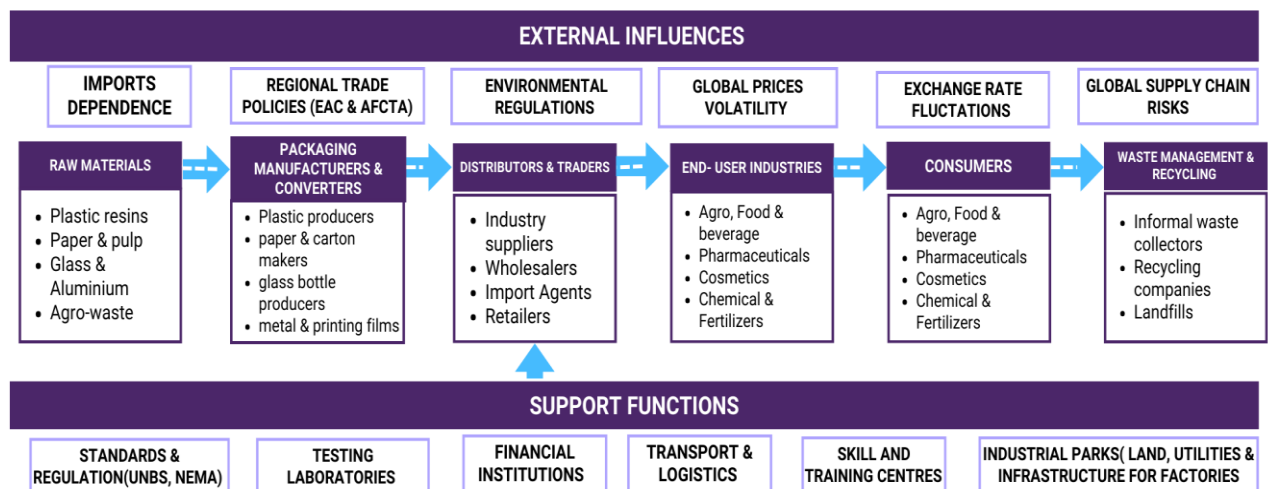
Value Chain Stage	Type of Actors	Examples Identified in the Mapping	Role in the Packaging System
Plastics	Flexible and rigid plastic packaging producers	Luuka Plastic Ltd, Nile Plastic Industries Ltd, Blowplast Uganda Ltd, Smart Pack (U) Ltd, Quality Plastics (U) Ltd	Manufacture plastic bottles, containers, films, and bags used across multiple industries.
Paper Based	Corrugated box and paperboard packaging manufacturers	Riley Packaging (U) Ltd, Redcorr (U) Ltd, East African Packaging Solutions, Cardboard Packaging Uganda Ltd, Linton Box Packaging Ltd	Produce cartons, corrugated boxes, and paper packaging mainly used by agro-processors and exporters.
Metal	Metal container manufacturers	Crown and Packaging Ltd, Jenvisen Packing Uganda Ltd	Produce metal cans and packaging for food and industrial products.
Glass	Glass container suppliers and distributors	East Africa Glassware Mart	Supply glass bottles and jars mainly used in food and cosmetics sectors.
Printing and Labeling	Graphic design, label printing, and packaging branding	Semliki Group of Companies, Uneeco Uganda Ltd, Graphic	Provide labels, packaging artwork, and regulatory product

	services	Systems Ltd	information.
Product Manufacturers / Packaging Users	MSMEs and large manufacturers	Food processors, coffee exporters, cosmetic producers	Use packaging to protect, transport, and market their products.
Waste Management and Recycling	Recycling companies and waste collectors	Aquila Recycling, Dado Recycle, Ecoplastile Ltd, Yo Waste	Collect and recycle used packaging materials such as plastics and paper.
Testing and Certification	Laboratories and technical testing institutions	UNBS, UIRI, Bureau Veritas, Intertek, Chemiphar, SGS	Conduct testing and certification for packaging materials and food safety compliance.
Regulatory Bodies	Government regulatory bodies	NEMA, National Drug Authority (NDA), Ministry of Trade Industry & Cooperatives	Regulate environmental compliance, product safety, and industrial development.
Industry Coordination	Industry associations and support organizations	IPPU, UMA, USSIA, KACITA, UEPB	Provide advocacy, industry coordination, and support services to manufacturers and MSMEs.

Table 3: Packaging Industry Distribution by Material Type

Packaging Type	Approximate Presence in Mapping	Key Observation
Plastic Packaging (Rigid & Flexible)	High	Dominates Uganda's packaging production sector
Paper-Based Packaging	Moderate	Growing due to sustainability and export demand
Metal Packaging	Low	Limited number of producers
Glass Packaging	Low	Mainly import-based with few manufacturers
Label Printing & Branding	Moderate	Essential support services for product packaging
Recycling & Waste Management	Growing	Driven by private recyclers and informal sector

STRUCTURE OF UGANDA'S PACKAGING VALUE CHAIN



5.1 Paper

Uganda's packaging raw material base is only partly developed. Some packaging materials are processed locally, but most key raw materials are imported. This shows that Uganda does not yet have strong upstream industries to support full local production.

In the paper and paperboard segment, Uganda does not have a large-scale pulp and paper factory. Even though the country has forest plantations that could support paper production, industrial pulp processing has not been established. Because of this, important materials such as kraft paper, folding boxboard, duplex board, food-grade paper, and tea filter paper are mostly imported from Kenya, India, China, and the Middle East. Recycling helps supply some paper materials, but poor waste sorting systems affect quality and consistency. This increases costs and keeps the country dependent on imports. In 2024, Uganda imported around US \$175 million of paper and paperboard. Imports from Slovakia alone were US \$4.34 million, including uncoated kraft paper and other paperboard used for packaging. Exports of paper packaging are low, only about US \$7.65 million in 2023. This shows that Uganda relies heavily on imported paper products for packaging.

5.2 Plastics

Uganda has a working plastic conversion industry. Local companies produce PET bottles, pouches, sachets, films, and rigid plastic containers used for food, drinks, and household products. However, the raw materials needed to make these products such as polyethylene (PE), polypropylene (PP), and PET resin are not produced in Uganda. These plastic resins are imported mainly from Saudi Arabia, India, China, and Kenya because Uganda does not have a petrochemical industry.

This heavy dependence on imports exposes the sector to exchange rate changes, global oil price increases, and high transport costs, especially since Uganda is landlocked. These factors raise production costs and affect competitiveness. Despite these challenges, companies such as Luuka Plastics, Graphics Systems, Supreme Flexibond, and East African Packaging Solutions continue to operate at different scales and supply packaging to local and regional markets. In 2024, the country spent about US \$635 million on these products. Between April 2024 and March 2025, there were 40 shipments involving 22 buyers and 19 foreign suppliers. Kenya supplied 93% of the shipments, and Kenya plus India supplied all of them. Overall, there were 887 shipments of plastic materials recorded in the latest tracking period, showing that imports happen very often. The plastics imported include polymer resins and semi-finished film products, which local companies use to make packaging. Uganda also exports plastics, but only about US \$62 million in 2024. This shows that local production is small compared to imports.

5.3 Glass and Metal Packaging Inputs

Uganda has natural resources needed for glass production, including high-quality silica sand near Lake Victoria and feldspar in the Mutaka area of the Rwenzori region. However, the country does not have a large glass container manufacturing plant. As a result, beverage and food companies import most of their glass bottles and jars. In 2023, imports of glass containers and related products were about US \$22 million. The Uganda Investment Authority has identified glass manufacturing as a strong investment opportunity because it could reduce imports and create jobs.

Metal packaging materials are also fully imported. Tinplate, which is used to make cans for products such as powdered milk and processed foods, is not produced locally. Uganda lacks steel rolling and coating facilities needed to make tinplate. This increases reliance on foreign suppliers and exposes the sector to global price changes and supply disruptions.

5.4 MSME Users

Uganda has over 1.1 million MSMEs, which contribute about 90% of private sector production. This creates strong demand for affordable and accessible packaging.

Main users include street food vendors, agro-processors (such as coffee, tea, flour, and dried fruit producers), dairy processors, small cosmetic manufacturers, pharmaceutical manufacturers among others. These businesses need packaging that protects products, improves shelf life, and meets safety standards.

However, MSMEs face several challenges. Many packaging suppliers require high minimum order quantities, which small businesses cannot afford. Customized packaging is often expensive or unavailable in small volumes. Some MSMEs believe that good packaging is too costly, and many lack knowledge about packaging standards and specifications. As a result, compliance with quality standards is sometimes inconsistent. When local suppliers cannot meet their needs, export-oriented MSMEs may import packaging through brokers, which increases their costs.

The table below highlights a selection of MSMEs in Uganda across key sectors and the products they offer, showing the wide demand for packaging solutions.

Table 5: MSMEs and the product lines in Uganda

Sector	MSMEs in Uganda	Goods/Products
Food and Agri-business	AgroWays (U) Limited	Buys grains such as maize sorghum, non GMO soy bean and barley from farmers, stores and manages them, and sells to processors and beverage industries. Also produces cassava flour.
	Olam Agri	Products include grain and oilseeds, edible oils, rice, specialty grains and seeds, animal feed and protein,
	Reco Industries Limited	Manufacturers fortified maize flour and corn soya blend flour, sauces (tomato ketchup, chilli sauce, garlic and chilli sauce, hot sauce)
	Atiak Sugar Factory	Produces sugar from sugarcane
	Biyinzika Poultry	All kinds of poultry feed, veterinary products and poultry
	Gulu Agricultural Development Company	Production of cotton, sesame, maize sunflower oil
	Grainpulse Ltd	Coffee export, grain processing and storage(maize, barley, maize flour, cassava flour, sorghum)

	Mayuge Sugar Industries Limited	Producers of sugar
	Maketa Snacks Company	Manufacturing of snacks such as Daddies , Groundnuts
	Psalms Food Industries Limited	Production of snacks, including potato crisps, plantain crisps, and peanut products
	Zahra Food Industries Ltd	Manufacture of dried fruit snack range(pineapple, mango, pawpaw, banana etc.), healing tea, fresh produce for exports like mango, pineapple, avocado, banana, sweet plantain among others, cashew nuts, macadamia, peanuts, chia seed, vanilla, moringa powder, plant based meat, breakfast bar
Cosmetic	Samona Products Factory	Manufacturer of soaps, detergents and candles
	MOHCA Beauty & Skincare Ltd- sheacare	Raw unrefined sheabutter
	Wispro (U) Ltd.	Hair care products, sanitizers, antipersperant, skincare products, nail polish remover
	Bellazuri Limited	Skin care and make up products
	Livara Shea Butter Cosmetics	Hair care products
Pharmaceutical	Rene Industries Ltd	Tablets, capsules, oral suspensions
	Mu Pharma co. SMC Ltd	medicines, cosmetics, and medical supplies
	Kampala Pharmaceuticatal Industries 1996 Ltd	Human medicines and nutraceuticals
	Quality Chemical Industries Ltd - QCIL	Manufacturing medicines to fight malaria, HIV/AIDS, and Hepatitis.
Chemicals and fertilizer	Grainpulse Ltd	Fertilizer blend
	Modern Agrichem	manufactures and sells pesticide chemical which include Herbicide, Insecticide and Fungicide.
	Itracom Fertilizer	Organic fertilizers
	Kel chemicals Ltd	Water treatment products, acids, fertilizers

5.5 REGULATORS

Several government bodies regulate different parts of the packaging sector. These include the Ministry of Trade, Industry and Cooperatives, the National Environment Management Authority, and the Kampala Capital City Authority. Their roles cover industrial policy, waste management, environmental protection, import regulation, and the development of Extended Producer Responsibility (EPR) systems.

Although laws and policies exist, enforcement remains weak in some areas, especially in plastic waste management and environmental compliance.

5.6 ACADEMIC AND RESEARCH INSTITUTIONS

Several universities and research institutions support industrial and agricultural development in Uganda. These include Makerere University, Kyambogo University, the Uganda Industrial

Research Institute, and the National Agricultural Research Organisation. Their research focuses on agro-processing, material science, biobased materials, and food preservation. However, Uganda does not yet have strong specialized training in packaging engineering or advanced packaging technology. This limits innovation and technical upgrading in the sector.

5.7 ASSOCIATIONS AND INDUSTRY GROUPS

Private sector coordination is supported by organizations such as the Uganda Manufacturers Association, the Private Sector Foundation Uganda, and the Uganda Small Scale Industries Association. These bodies support advocacy, policy dialogue, standards awareness, and MSME training.

The emerging Institute of Packaging Partners Uganda (IPPU) also aims to bring packaging stakeholders together to improve knowledge sharing and sector coordination. However, the packaging sector still lacks a strong, unified industry body focused only on packaging development.

5.8 WASTE MANAGEMENT

Waste management in Uganda is primarily the responsibility of local governments. Each city or municipality manages waste collection, transportation, and disposal within its jurisdiction. In Kampala, the Kampala Capital City Authority (KCCA) leads waste management operations, while other cities such as Jinja, Gulu, Mbale, and Soroti operate their own systems, often facing financial and technical constraints. At the national level, policy and regulatory oversight is provided by institutions including the Ministry of Local Government, Ministry of Lands, Housing and Urban Development, National Environment Management Authority (NEMA), and the Ministry of Water and Environment. However, overlapping mandates sometimes weaken coordination.

Public-private partnerships are commonly used for waste collection, with private operators charging households between UGX 3,000 and 30,000 per month. These fees can be unaffordable for low-income households, leading to low service coverage. Kampala generates approximately 2,500 tons of waste per day, yet less than half is properly collected, and collection rates are even lower in secondary cities. Most waste is disposed of at the Kiteezi Landfill, which has exceeded its capacity and lacks adequate sanitary infrastructure.

The informal sector plays a critical role in recycling. Waste pickers, scrap dealers, and community collectors recover plastics, metals, and paper, although overall recycling rates remain low, with only about 8% of plastics recovered. Some private sector initiatives, including partnerships with companies such as Coca-Cola Beverages Uganda, are helping establish collection systems and support recycling. In addition, the government is advancing policies such as Extended Producer Responsibility (EPR) to encourage producers and importers to contribute to packaging waste management and recycling.

5.9 QUESTIONNAIRE FINDINGS

The tables below present a consolidated overview of responses from stakeholders across Uganda's packaging value chain, capturing their roles, capacities, challenges, and areas for strategic intervention

Table 6: Packaging Materials & Technology

Packaging Material	Stakeholders Producing	Challenges
Paper & Paperboard	Manufacturers, MSMEs	Limited local supply of kraft and white kraft liner; machinery limited to one size
Plastics	Manufacturers, MSMEs	Imports dominate; limited sustainability options
Glass	Manufacturers	Moderate adoption; limited local recycling
Metal	Manufacturers	Limited local suppliers

Flexible Packaging	MSMEs, Manufacturers	Technology gaps; imported machines required for variety
Biodegradable Materials	Few SMEs, NGOs	High cost, limited availability
Other	N/A	Innovation limited

Table 7: Testing & Certification Capacity

Certification	Stakeholders	Adoption Level	Observations
Physical Testing (strength, compression)	Few	Moderate	Available, but limited coverage
Chemical Migration Testing	Few	Low	Rarely available locally
Microbiological Testing	Few	Very Low	Limited in-country capacity
Environmental Testing	Few	Low	Scarce infrastructure
ISO 9001	Many	Medium	Mostly larger firms
ISO 22000	Some	Low	Mostly food-related
ISO 14001 / GMP	Few	Low	Limited SMEs adoption
Other Certifications	Few	Very Low	Niche / specialized

Table 8: Sustainability Practices

Sustainable Practice	Adoption Level	Barriers
Use of recycled materials	Limited	Low availability, cost, lack of awareness
Use of biodegradable materials	Emerging	Limited local production, high import dependency
Waste reduction programs	Minimal	Low SME engagement
Extended Producer Responsibility (EPR)	Rare	Policy gaps, limited enforcement
None / no practice	Many SMEs	Cost constraints, lack of knowledge

Table 9. Skills & Training Gaps

Area of Skill Gap	Stakeholders Reporting	Observations
Packaging technology & design	Most SMEs	Circular economy, intelligent packaging, shelf-life extension technologies
Regulatory compliance & standards	Most stakeholders	Low knowledge of UNBS & EU PPWR standards
Recycling & waste management	Many SMEs	Minimal structured training locally
Machinery operation & automation	Few	Lack of adaptable paperbag machinery

Table 10: Regulatory Challenges

Regulatory Challenge	Impact	Notes
High compliance costs	High	High Burden for SMEs
Limited clarity of standards	High	UNBS standards exist but poorly enforced

Delays in certification	Medium	Process inefficiencies
Overlapping mandates	Medium	Confusion in roles between regulators
Limited testing infrastructure	High	Barrier to export and compliance
Import dependency	High	Local material scarcity
Other	Medium	e.g., lack of policies for SMEs, polythene ban enforcement

Table 11: Investment Gaps

Investment Gap	Observations
Local production of kraft & white kraft liner	Limited supply affects variety of paper packaging
Machinery for variable/adjustable paperbags	Current machines limited to one size/type
Testing facilities for package integrity	Few labs available locally
Recycling infrastructure	Minimal; few operational recycling initiatives
Access to funding & financing	SMEs struggle to invest in modern machinery & materials
Digitalization of supply chain	Low adoption; manual systems prevail
Sustainable material supply	Reliance on imports from India, China, Europe, Singapore, Kenya

6.0 IMPORT VS. LOCAL PRODUCTION DYNAMICS

Uganda's packaging sector relies heavily on imported raw materials. The country exports only small amounts of packaging materials. Paper cartons, boxes, and plastic packaging products together make up a small share of exports, with plastic packaging mainly going to DR Congo, South Sudan, and Rwanda. Overall, export values are much lower than import values, showing that local production cannot meet demand.

Uganda does not have large petrochemical plants, pulp and paper mills, or tinsplate factories to supply these materials locally. Even though the country has resources like forest plantations and silica sand, they have not yet been fully developed for industrial use. Because of this, the sector is vulnerable to changes in exchange rates and global price increases, especially for oil-based plastics. This also leads to high foreign exchange spending. In some cases, locally produced packaging is more expensive than imported products from countries like Kenya, where industries are more advanced and operate at a larger scale.

TABLE 12: TRADE DATA FOR PACKAGING RAW MATERIALS IN UGANDA

Material Type	Import Value (USD)	Key Suppliers	Notes on Imports	Export Value (USD)	Key Export Destinations	Notes on Exports
Plastics & Plastic Packaging	635.27 million (2024)	Kenya (93% of shipments), India	High number of shipments (887), strong domestic demand for polymer resins & film products	62 million (2024)	DR Congo, South Sudan, Rwanda	Exports are small compared to imports, showing low local production
Paper & Paperboard	175.35 million (2024)	Slovakia, other EU countries	Used for cartons, boxes, bags; imports include kraft paper & paperboard	7.65 million (2023)	Regional markets	Limited export capacity, small fraction of national exports
Glass & Glassware	22 million (2023)	Mainly imported (no local production)	Bottles, jars, containers for food & beverages	Minimal	-	Dependence on imports highlights lack of domestic glass manufacturing

7.0 CAPACITY ASSESSMENT OF THE PACKAGING VALUE CHAIN IN UGANDA

The packaging sector in Uganda is a vital component of the country's manufacturing and export industries. Despite the growing importance of packaging in Uganda's manufacturing and agro-processing sectors, several structural constraints continue to limit the sector's competitiveness and growth. The assessment of its capacity reveals several delicate areas that require targeted interventions to address and enjoy their full potential and ensure sustainable growth.

1. Capacity for Packaging Design, Innovation and high dependence on imported raw materials:

Uganda's packaging industry currently lacks a strong ecosystem for design and technological innovation. Most packaging products are produced using traditional methods with minimal focus on creativity, functionality, or sustainability. This limitation stems from inadequate investment in research and development (R&D) and a scarcity of local designers and innovators with expertise in packaging engineering. Consequently, manufacturers often rely on imported packaging solutions, which are costly. Uganda relies heavily on imported inputs such as plastic resins, paperboard, inks, adhesives, and aluminum sheets. This dependence exposes the sector to foreign exchange fluctuations, supply disruptions, and higher production costs. The absence of domestic pulp and paper mills and petrochemical industries further constrains the development of local packaging manufacturing. The absence of innovation hinders the sector's ability to develop eco-friendly, cost-effective, and consumer-friendly packaging options, thereby affecting competitiveness both locally, regionally and in export markets.

2. High logistic costs: Logistics remains a persistent bottleneck for packaging supply chains across the continent, with unreliable infrastructure and fragmented service providers often disrupting material flows. The lack of sufficient infrastructure hampers the efficiency and scalability of the packaging value chain in Uganda. Raw materials such as plastics, paper, and cardboard are either imported or sourced from limited local suppliers, leading to supply chain vulnerabilities and high loss and waste. Transportation infrastructure, including roads, resulting into high logistics costs, delays, and increased lead times for packaging manufacturers. These infrastructural gaps increase production costs and reduce the sector's ability to meet increasing demand efficiently.

3. High Energy Costs: Energy costs in Uganda remain high relative to many neighboring countries, primarily due to reliance on expensive and unreliable electricity supplies. The cost of energy is likely to increase in the face of the political instabilities in the world. For manufacturing processes such as plastic molding, printing, and coating, consistent and affordable power is essential. The high energy prices inflate operational costs for local packaging producers, making their products locally highly costed, less competitive in regional and international markets. This drawbacks are hurdles investment in energy-intensive packaging technologies and limits the capacity for scaling up production.

4. Limited Access to Finance and Modern Technology: Access to affordable finance is a significant obstacle for packaging manufacturers in Uganda. Many MSMEs in the sector struggle to secure loans or investment due to unbearable collateral requirements and limited banking sector awareness of the sector's potential. This financial constraint impacts negatively on investments in modern machinery, technology upgrades, and capacity-building undertakings.. Many packaging converters, particularly MSMEs, operate with basic machinery and limited automation. Access to modern technologies such as flexographic printing, high-resolution digital printing, automated converting equipment, and advanced finishing systems remains limited due to high capital investment requirements. Currently many firms operate below their optimal production capacity, unable to upgrade their production lines or adopt innovative packaging practices that could improve quality and reduce costs. This affects product quality, productivity, and the ability to meet international packaging standards

5. Skills Technical knowledge and Training Gap: The sector suffers from a skills gap, characterized by lack of trained personnel with expertise in packaging design, manufacturing processes, quality control, and environmental management. Most workers acquire skills informally or on job training programs. There is a shortage of specialized technical skills in packaging design, packaging engineering, materials science, and packaging technology. Many enterprises lack expertise in areas such as sustainable packaging design, packaging optimization, and export-compliant labeling. As a result, businesses often rely on trial-and-error approaches rather than structured packaging development processes. This skills deficit results in limited

innovation, and an inability to meet international standards. It also affects the sector's capacity to use sustainable practices, such as eco-friendly packaging solutions, which require specialized knowledge and technical expertise.

Fragmented Regulatory Environment: The regulatory framework governing packaging involves multiple institutions responsible for standards, environmental protection, and industrial development. This is one of the primary challenges, the fragmentation of regulations governing packaging. Multiple agencies such as the Uganda National Bureau of Standards (UNBS), the National Environment Management Authority (NEMA), and the Uganda Revenue Authority (URA) have overlapping mandates related to standards, environmental compliance, and taxation. This fragmentation creates confusion among MSMEs about which regulations take precedence, leading to overlaps, inconsistencies, and gaps in enforcement. For MSMEs, navigating regulatory requirements can be complex and time-consuming, particularly when compliance involves multiple agencies and technical standards. Furthermore, majority of regulations are not tailored specifically to the packaging sector. There is a lack of sector-specific standards that address modern packaging innovations, environmental considerations such as biodegradable or recyclable packaging, and export requirements. This gap affects the manufacturers' ability to innovate and comply with international best practices, especially as global markets intensify standards on sustainable packaging.

Limited Testing and Quality Assurance Infrastructure

Uganda's current laboratory infrastructure for packaging material testing is underdeveloped and concentrated in a few government or tertiary institutions. Uganda has a limited number of laboratories capable of conducting advanced packaging material testing, food contact safety verification, and packaging performance assessments. Many of these laboratories lack the state-of-the-art equipment needed to perform required analyses, such as chemical composition, barrier properties, biodegradability, and safety assessments. Without access to advanced testing facilities, manufacturers cannot accurately evaluate the quality, safety, and environmental impact of their packaging products. Limited laboratory capacity hinders the ability of MSMEs to verify compliance with standards set by Uganda National Bureau of Standards (UNBS) and other region or international organizations. This restricts the ability of MSMEs to certify packaging materials for export markets and increases reliance on external testing services. This reliance on external laboratories located outside the country leads to increased costs, longer turnaround times, and logistical challenges. In addition, the technical capacity of countries' laboratories is wanting to keep pace with evolving packaging industry and technological advancements.

Weak Recycling and Circular Economy Systems

Uganda's recycling infrastructure remains underdeveloped, with limited facilities for collecting, sorting, and processing recyclable packaging materials such as plastics, paper, and metals. Many recyclable materials are disposed of in landfills or informal dumpsites, leading to environmental pollution. Although recycling activities exist, they are largely driven by private companies and informal waste collectors. Infrastructure for large-scale collection, sorting, and processing of packaging waste remains insufficient. This limits the country's ability to transition toward circular packaging systems and manage increasing volumes of packaging waste. There is a lack of implementation of policies that promote recycling and circular economy principles within the packaging sector. This results in minimal incentives to get involved in recycling activities.

8.0 Testing, Standards Compliance, Export readiness (including PPWR awareness), Skills, and Coordination mechanisms (MSMEs and Export-oriented sectors)

1. Access to Testing Facilities

The Testing situation in Uganda is characterised with limited local laboratories testing abilities to undertake packaging material analysis. Most MSMEs are not able to access affordable and accredited testing services. The mandate of testing goes mainly to UNBS laboratories, but they lack capacity hence other private laboratories close the gap but at prohibitive cost. Furthermore, the MSMEs are unaware of packaging testing requirements. Testing facilities are vital for ensuring packaging materials meet safety, quality, and environmental standards. Insufficient testing capacity affects export compliance, more especially with international standards requiring specific and rigorous packaging certifications.

2. Standards Compliance

The Uganda National Bureau of Standards (UNBS) sets standards governing the packaging, but awareness and enforcement are low, particularly among MSMEs. Compliance to national regional and international packaging standards is the ticket for market access. Currently this low awareness and limited knowledge of standards and certification processes among the MSMEs. The UNBS is mandated to provide technical guidance and support services, but it lacks the capacity to undertake on this task. Outsourcing such services is costly hence capacity barriers to achieving compliance. This situation impacts negatively on exports leading to rejection at borders and loss of export markets.

3. Export Readiness & PPWR (Extended Producer Responsibility) Awareness

National Environment Act No.5 2019 introduces provisions on extended producer responsibility so that producers of any product have the duty to manage the design and disposal of the product and monitor its lifecycle in order to deliver the best overall environmental outcome. This indicated the Uganda to meet international packaging regulations, including PPWR (Plastic Packaging Waste Regulations) and other environmental standards.

However, awareness of PPWR and similar regulations is not widespread among the MSMEs and exporters. Most of the MSMEs are unaware of their responsibilities or how to adapt their packaging practices. This has been constrained by low and limited knowledge and capacity to implement sustainable packaging practices. This is further aggravated by lack of access to affordable eco-friendly packaging options combined with inadequate information on export standards and compliance procedures. The most likely outcome of such situation is reduced export competitiveness and potential trade barriers due to non-compliance with international environmental standards.

4. Skills and Capacity

Capacity building for development of skilled personnel is an ingredient for designing compliant, sustainable, and innovative packaging solutions. Currently there is skills gap in the packaging sector with respect to packaging design, testing, and sustainability practices among MSMEs. There is a need to develop technical training programs tailored for MSMEs and development of specialized knowledge in eco-design and sustainability. Development of a specific curriculum address the different aspects of the packaging sector can alleviate the problem. However, lack of skills limits the MSMEs to innovate and meet global packaging standards.

5. Coordination Mechanisms

For effective coordination among government agencies, industry players, and development partners to ensure a cohesive packaging sector will require concert effort to promote stakeholder engagement, in the packaging sector more especially among the MSMEs. To achieve this will require to deal with the weak or nonexistence of communication channels and information sharing among the MSMEs. This calls for the establishment of innovation platforms for MSMEs to learn, share, access support and resources plus sharing information. Furthermore, there is need of more efforts for integration of packaging sector development within trade and environmental policies.

The table below summarises the missed opportunities for capacity building, resource mobilization, and policy implementation.

Table 13: Missed opportunities for capacity building, resource mobilization, and policy implementation.

Aspect	Current Status	MSME Constraints	Impact on Export-Oriented Sectors
Access to Testing	Limited local labs; expensive, inaccessible testing services	High Cost, low awareness, logistical barriers	Export rejections, delayed certification.
Standards Compliance	Low enforcement; low awareness of standards and certification among MSMEs	Limited knowledge, certification process and high costs involved.	Market access barriers, loss of export opportunities
Export Readiness & PPWR Awareness	Low awareness of environmental regulations and standards	Limited information, awareness, capacity gaps in PPWR	Non-compliance risks, trade barriers
Skills & Capacity	Skills gap in design, testing, and sustainability practices	Limited technical training programs, technical expertise support	Reduced innovation, competitiveness
Coordination Mechanisms	Fragmented stakeholder engagement	Lack of platforms, poor communication Channels	Inefficient policy implementation, resource gaps

9.0 STRATEGIC OPPORTUNITIES FOR STRENGTHENING THE PACKAGING SECTOR

Industrial development cannot take off without a vibrant packaging sector. The packaging sector in Uganda plays a vital role in enhancing the quality, safety, and marketability of goods across various industries, including agriculture, manufacturing, and retail. As the economy grows and consumer preferences and taste change, there are tremendous strategic opportunities to strengthen this sector. Strengthening the packaging sector will support industrial development and also contributes to broader economic growth and sustainable development goals in the country. The areas of focus are underline below:

- **Growing Demand from Agro-Processing and Manufacturing**

Uganda's policy focus on agro-industrialization and value addition is expected to increase demand for packaging solutions. The expanding sectors of coffee processing, food manufacturing, cosmetics, and pharmaceuticals requires improved packaging to meet both domestic and export market requirements.

- **Development of Paper-Based and Sustainable Packaging**

Global shifts toward environmentally sustainable packaging create opportunities for Uganda to develop paper-based and biodegradable packaging solutions. With appropriate investment, the country could expand its production of paper bags, corrugated packaging, and fiber-based packaging materials that support both domestic industries and export markets.

- **Expansion of Regional Packaging Markets**

Uganda's strategic location within the East African Community (EAC) provides opportunities for packaging manufacturers to supply regional markets. As trade integration deepens within the region, demand for standardized and export-compliant packaging is expected to increase.

- **Investment in Packaging Technology and Innovation**

Introducing modern converting technologies, digital printing systems, and automated packaging machinery could significantly improve product quality, efficiency, and competitiveness. Technology upgrades would enable local manufacturers to produce higher-value packaging products and compete with imported packaging materials.

- **Strengthening Packaging Skills and Training Systems**

Developing specialized training programmes in packaging technology, design, and materials science could help address existing skills gaps. Packaging training centres and industry partnerships could play an important role in building technical capacity across the sector.

- **Development of Circular Packaging Systems**

Expanding recycling infrastructure and implementing circular economy approaches, such as Extended Producer Responsibility (EPR), could improve the management of packaging waste while creating new economic opportunities in recycling and secondary materials production.

10.0 SWOT ANALYSIS OF UGANDA'S PACKAGING VALUE CHAIN

TABLE 14: SWOT Analysis of Uganda's Packaging Value Chain

Strength	Weakness
Growing packaging conversion sector, especially in plastics and paper-based materials.	Heavy dependence on imported raw materials such as plastic resins, tinplate, and pulp and paper.
Established converters producing flexible films, PET bottles, cartons, labels, and rigid packaging.	Exposure to exchange rate risks and global price volatility due to import dependence.
Strong demand driven by agro-processing, MSMEs, and the food and beverage sector	Technology gaps; advanced packaging materials often imported.
Strategic location within the East African Community, providing regional market access.	Many small converters use outdated equipment

Access to continental markets through the African Continental Free Trade Area.	Weak and poorly coordinated recycling systems
Natural resources such as silica sand and agricultural residues for potential upstream and biodegradable packaging production.	Fragmented implementation and limited coordination of existing packaging policies and strategies
Active informal recycling sector recovering plastics, paper, and metals.	Skills gaps in packaging engineering, materials science, and quality assurance
Regulatory oversight and certification provided by the Uganda National Bureau of Standards.	Limited technical guidance for advanced packaging applications.
Opportunities	Threats
Development of upstream industries such as glass manufacturing and agro-based biodegradable packaging.	Global oil price fluctuations affecting plastic resin costs.
Implementation of Extended Producer Responsibility (EPR) systems by the National Environment Management Authority to strengthen recycling and create green jobs.	Currency depreciation is increasing the cost of imported raw materials.
Growing global demand for sustainable and recyclable packaging solutions.	Stronger regional competitors, particularly from Kenya, dominating higher-value segments.
Industrial park development and cluster-based production to reduce costs and improve scale.	Potential tightening of environmental regulations increasing compliance costs.
Improved regional trade under continental agreements increasing export potential.	Climate change pressures leading to restrictions on conventional plastics.
Innovation in biodegradable packaging using local agricultural residues.	Weak enforcement of standards and informal competition undermining quality.

11.0 National Priorities and Recommendations

Based on Uganda's National Packaging and Branding Strategy and UNIDO programming, the following priority intervention areas and practical recommendations have been identified:

1. Waste Management:

In Uganda, packaging waste, particularly plastics and single-use materials, has become a menace to the environment due to inadequate waste collection and limited disposal infrastructure. The experienced flooding in cities has been attributed to the blocking of water drainages with packaging waste. Improving waste management calls for investment in modern packaging waste collection systems, establishing and supporting recycling centers, and promoting community-based packaging waste collection initiatives. Recycling can reduce the volume of packaging waste in landfills and the environment, while also creating opportunities for the development of recycling industries. Promoting sustainable practices among manufacturers and consumers is essential to minimize environmental impact and support Uganda's commitment to environmental conservation.

2. Circular Economy

Adopting circular economy principles in Uganda's packaging sector encourages designing packaging products that are recyclable, reusable, or biodegradable. This approach helps reduce waste generation and conserves natural resources. For example, encouraging local manufacturers to develop packaging made from biodegradable materials locally available or designing packaging that can be reused multiple times aligns with Uganda's environmental goals. Implementing policies and incentives to support circular practices can facilitate a transition toward a more sustainable eco-friendly packaging industry, promoting innovation and reducing dependence on imported packaging materials.

3. Quality and Safety:

Strengthening regulatory frameworks in Uganda with respect to standards for packaging materials is important in ensuring their safe, durable, and environmentally friendly. Strengthening enforcement mechanisms ensures compliance among manufacturers and importers. This can include regular inspections, certification processes, and penalties for non-compliance. This will align Uganda with international best practices, more importantly meeting export markets demand higher standards for packaging.

4. Skills Development:

Building capacity within Uganda's packaging sector requires targeted training programs for designers, manufacturers, and waste management professionals. This includes technical skills in sustainable packaging design, manufacturing processes, and waste handling. Introduction of packaging training program in vocational training centers, collaborating with higher educational institutions can help develop a skilled technical workforce to manage modern packaging technologies.

5. Research for Development (R4D)

Supporting R4D in Uganda's packaging sector involves investing in research institutions and industry partnerships to develop new sustainable materials, innovative packaging designs, and efficient manufacturing technologies. Research could focus areas on the use of biopackaging material available such as cassava, rice and banana fibers, in the country to produce biodegradable plastics, lightweight packaging, and smart packaging solutions that improve product safety and shelf life. The R4D efforts in packaging can position Uganda as a centres of excellence for packaging innovation in the region.

Recommendations

1. Establish a National Packaging Authority:

To effectively coordinate the development of Uganda's packaging sector, the government should establish a National Packaging Authority (NPA). This authority would oversee the implementation of the National Packaging and Branding Strategy, ensuring alignment with national development goals. The NPA would regulate packaging standards, promote sustainable packaging practices, and facilitate the adoption of innovative packaging solutions. It would also serve as a platform for stakeholder engagement, including industry players, government agencies, and development partners like UNIDO, to support the sector.

2. Implement a comprehensive Extended Producer Responsibility (EPR) policy:

Uganda has in place the EPR policy that makes producers responsible for the lifecycle management of their packaging waste. This policy encourages manufacturers to design packaging that is easier to recycle, reuse, or biodegrade, while also establishing collection

and recycling systems. However it has limited awareness among the packaging stakeholders.

3. Develop a packaging waste management plan

A national packaging waste management plan should be developed to address the entire waste lifecycle, from collection to disposal. This plan would involve mapping waste generation hotspots, establish waste collection points, and create sorting and recycling facilities. It should incorporate best practices from UNIDO's technical assistance programs, emphasizing environmentally sound and cost-effective waste management infrastructure. The plan would also promote public awareness campaigns to encourage responsible consumption and waste segregation at the source, ultimately reducing litter and environmental pollution.

4. Provide training and capacity-building programs

Capacity development programs should target designers, manufacturers, waste collectors, and recycling operators, emphasizing sustainable packaging design, waste handling, and recycling techniques. Collaboration with local technical institutes, industry associations, and UNIDO's expertise can facilitate knowledge transfer, improve technical skills, and promote innovation. Building a skilled workforce will enable Uganda to adopt modern, environmentally friendly packaging solutions and improve overall sector competitiveness.

5. Establish a waste management data collection system

Uganda needs a robust data collection system focused on packaging waste. Such a system would gather data on waste generation volumes, collection coverage, recycling rates, and disposal methods. Gathered data will help identify key waste streams, optimize collection routes, and evaluate the effectiveness of waste management initiatives, ultimately supporting evidence-based policymaking and resource allocation

6 Achieve a significant reduction in packaging waste

Uganda aims to cut packaging waste generation by 50%, aligning with the objectives of its National Packaging and Branding Strategy and supported by UNIDO's technical assistance. This can be achieved through promoting eco-design principles that minimize material use, encouraging reuse and refill systems. Encouraging consumers to adopt responsible consumption behaviors are also necessary. Fostering partnerships with waste management enterprises to improve collection and recycling infrastructure will reduce waste generation translating into tangible and visible environmental benefits.

7. Increase the use of locally sustainable packaging materials

The goal is to ensure that at least 50% of all packaging materials used in Uganda are sustainable—biodegradable, compostable, or made from recycled content. This involves incentivizing local manufacturers to adopt environmentally friendly materials such as bioplastics, paper-based packaging, and other renewable resources available in the country. Policies promoting research for development into locally sourced sustainable materials, coupled with certification schemes for eco-friendly packaging, can accelerate the process. UNIDO's programming can support technology transfer and capacity-building for sustainable material production, helping Uganda reduce reliance on imported plastics and foster a green packaging industry.

8. Develop a packaging industry that is globally competitive

Uganda dreams to develop a packaging sector characterized by innovation, quality, and sustainability, capable of competing in international markets. This entails investing in modern manufacturing technologies, establishing quality assurance systems, and fostering innovation hubs for packaging design. Developing export-oriented packaging

solutions that meet international standards will open new markets and attract foreign investment. UNIDO's technical support can facilitate access to global best practices, standards, and certification processes, positioning Uganda as a regional leader in innovative and sustainable packaging solutions.

9. Enhance the capabilities of packaging professionals

A skilled personnel a key ingredient for advancing Uganda's packaging sector. There is need for training and capacity-building programs with a focus on packaging designers, manufacturers, quality control personnel, and waste management professionals. These programs can focus on sustainable design, eco-labeling, quality assurance, innovation in smart packaging and recycling technologies. Collaborations with technical colleges, universities such as Makerere University, and UNIDO's experts can facilitate knowledge exchange through tailored training programs and certifications. Capacity building is ne to keep pace with to changing global best practices.

10 Establish Uganda as a regional leader in packaging

In the long run , Uganda should position itself as regional hub for innovative, sustainable, and efficient packaging solutions in the EAC region. Promoting Uganda's success stories and fostering partnerships with neighboring countries can establish a reputation for excellence in sustainable packaging. This can involve hosting regional packaging forums, trade fairs, and innovation in packaging competitions. Developing a strong ecosystem of industry players, academia, and government agencies will boost Uganda's status as a leader in packaging, driving economic growth and environmental sustainability in the region.

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ANNEX ONE

Uganda Packaging Value Chain Stakeholder Questionnaire

Background of the Study

Packaging is a critical enabler of food safety, export competitiveness, branding and sustainability in Uganda. Recent initiatives supported by UNIDO, including the WPO Masterclass in Packaging Technology (August 2025) and the Packaging for SMEs Workshop (October 2025), confirmed both the strategic importance of packaging and the presence of significant gaps across the packaging value chain.

Stakeholders highlighted fragmentation among actors, limited access to testing and standards compliance services, capacity constraints among MSMEs, and low awareness of emerging international requirements such as the EU Packaging and Packaging Waste Regulation (PPWR). In this context, a structured mapping and capacity assessment of the packaging value chain is required to inform targeted technical assistance and future programme design.

The objective of the assignment is to map the packaging value chain stakeholders in Uganda and assess their capacities, in order to identify gaps, priorities and opportunities for coordinated capacity building, policy support and investment

Instructions

Please complete all sections. Some questions allow multiple responses. Estimated completion time: 15–20 minutes.

Section A: Institutional Profile

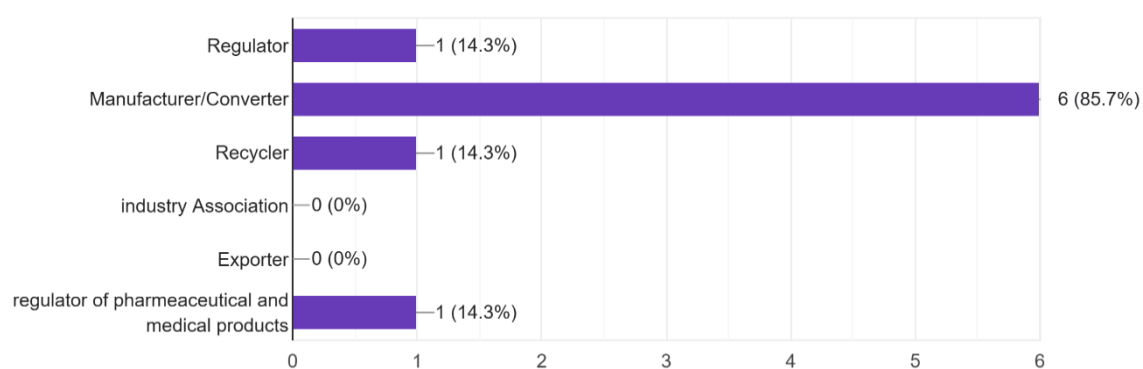
Section B: Packaging Role & Capacity

Section C: Trade & Regulatory Environment

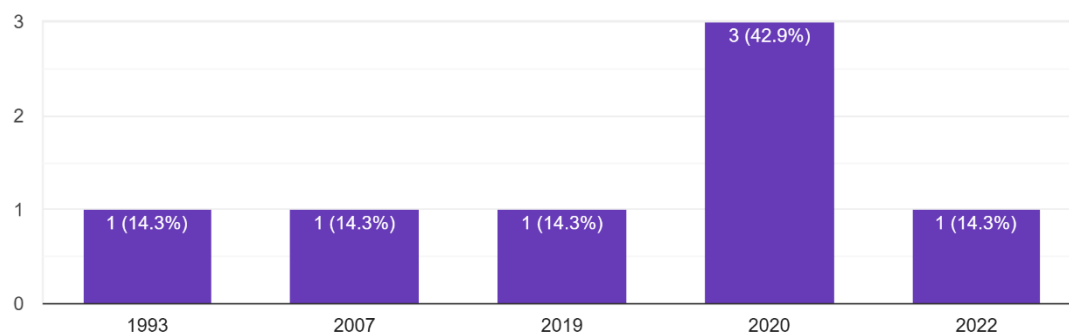
Section D: Sustainability & Circular Economy

Section E: Investment, Skills & Recommendations

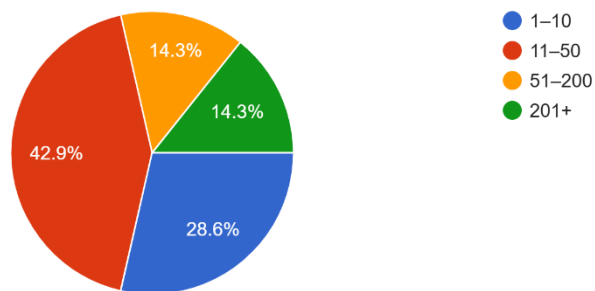
Type of stakeholders



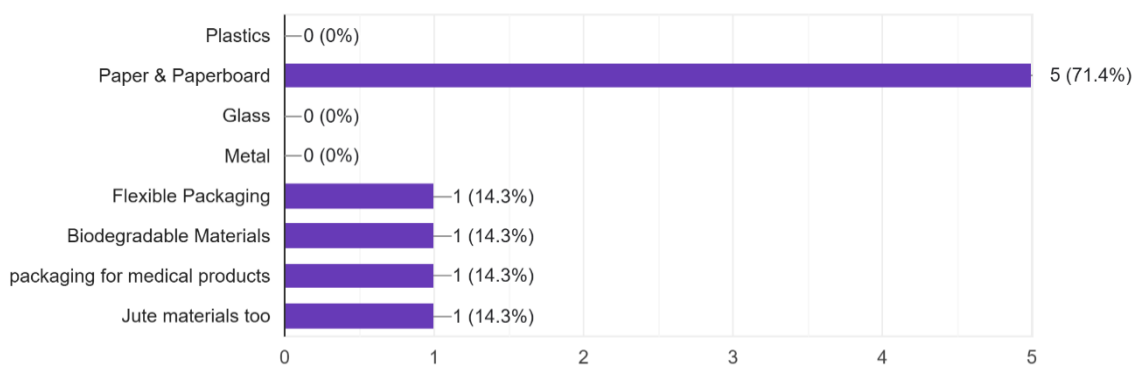
Year Established



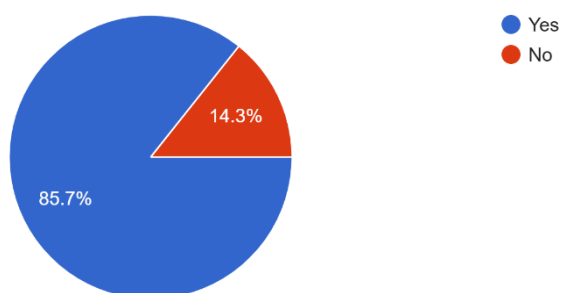
4. Number of Employees:



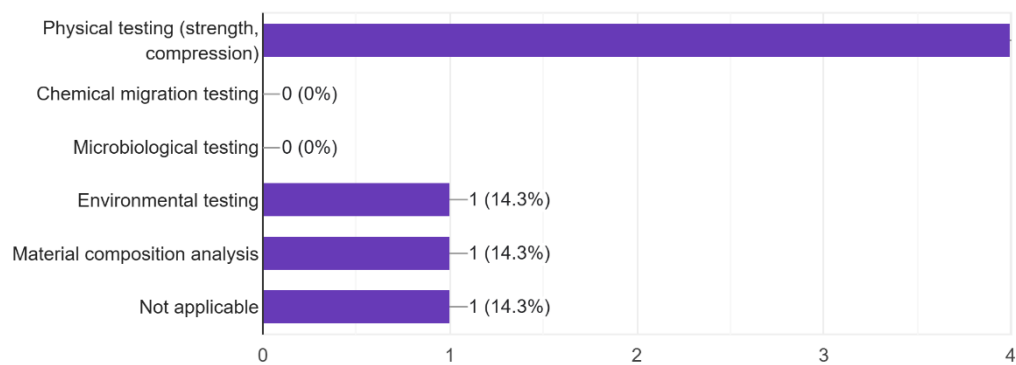
Which packaging materials do you regulate/produce/support? (Select all that apply)



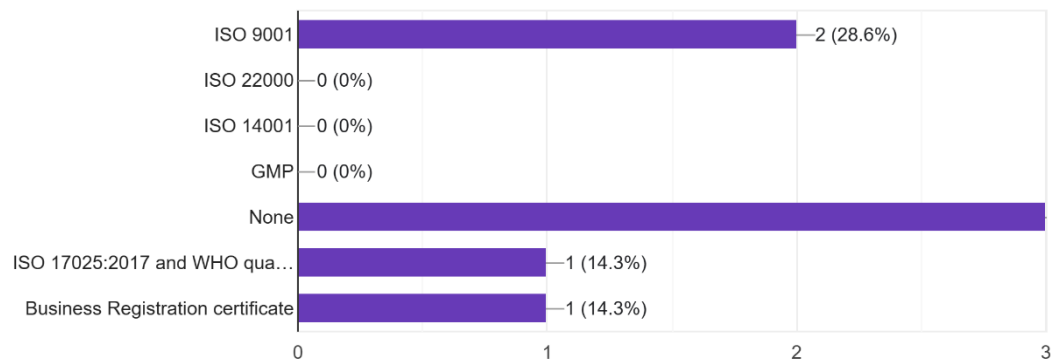
Do you operate or have access to packaging testing facilities?



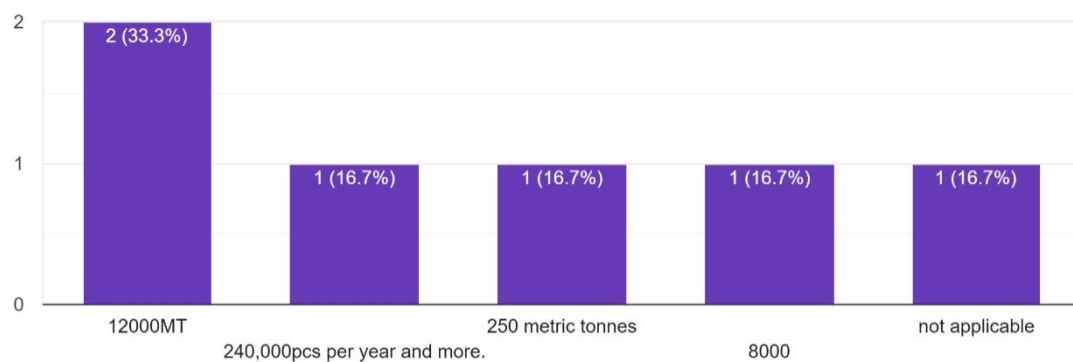
Which testing capabilities are available? ☐ Material composition analysis



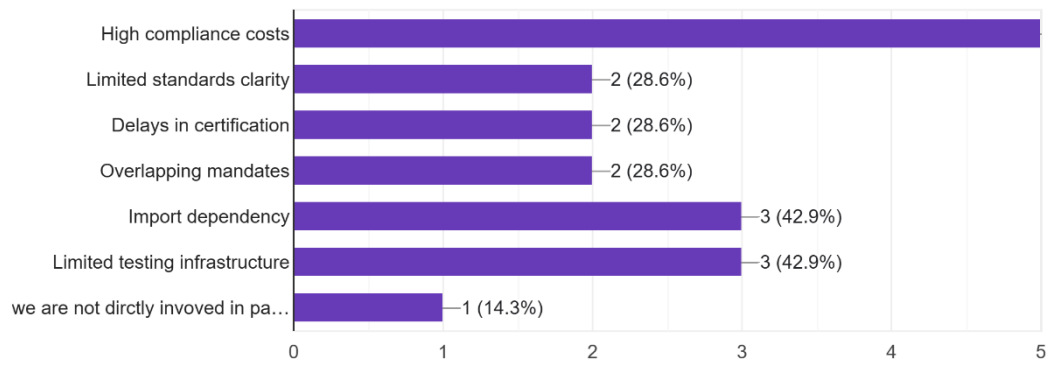
8. Do you hold any certifications?



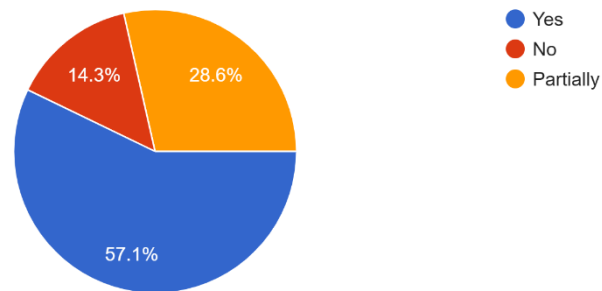
9. What is your estimated annual production or regulatory coverage capacity?



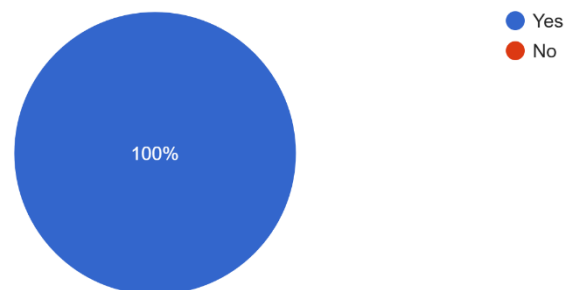
10. What are the main regulatory challenges affecting packaging operations?



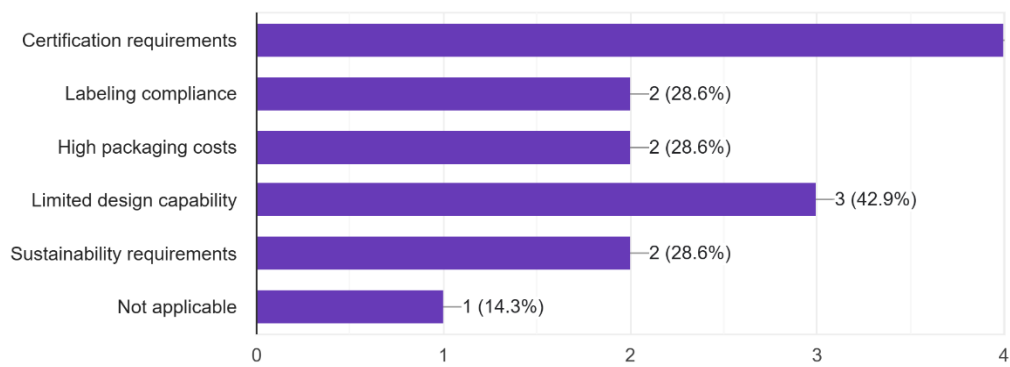
11. Are current packaging standards adequate?



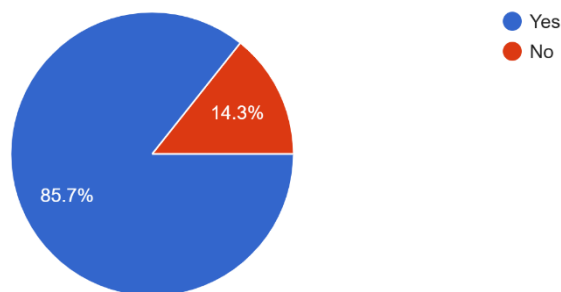
12. Does your organization engage in export markets?



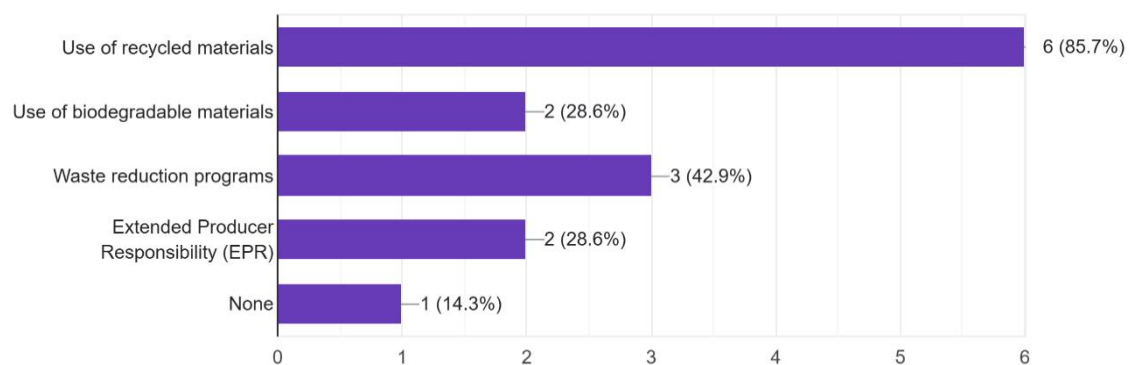
13. If yes, what packaging-related barriers affect exports? (Select all that apply)



14. Does your organization implement or support recycling initiatives?



15. Which sustainable packaging practices are in place?



16. What are the biggest barriers to sustainable packaging in Uganda? Limited availability of alternatives

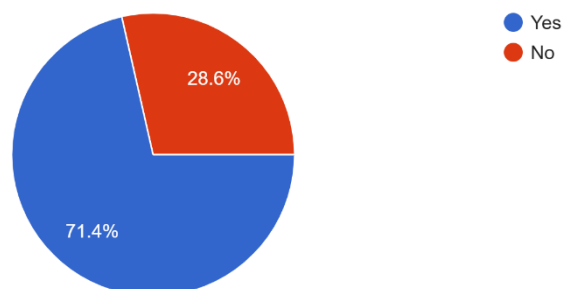
- Lack of awareness in sustainable packaging options

- reliance on imported materials, limited local capacity for specialized, sustainable, and high-standard packaging materials , cost of materials and limited packaging technology
- Limited capital
- No price regulations
- No laws govern production of paper bags for SMEs
- Imports from Chinese
- High taxes
- The sachet economy & affordability trap, Low consumer awareness and demand

17. What major investment gaps exist in Uganda's packaging sector? Paper manufacturers for kraft and white kraft liner

- Limited access to packaging funding
- manufacturing orderable and recyclable packaging; testing facilities for package integrity
- Recycling
- An investment gap in limited Materials supply here in Ug.. Materials are limited on market that can be used to make variety paper packaging items.
- Another investment gap is machinery that does adjustable paper bags.
- We have clients with different needs sizes or materials. Now by all means we have to stay using manual because if we import a machine that does paper bags with handles it will be fed with one paper kind.
- Not variety of papers and channeled to making one size not variety of sizes.
- High taxes on imported papers
- Limited technology

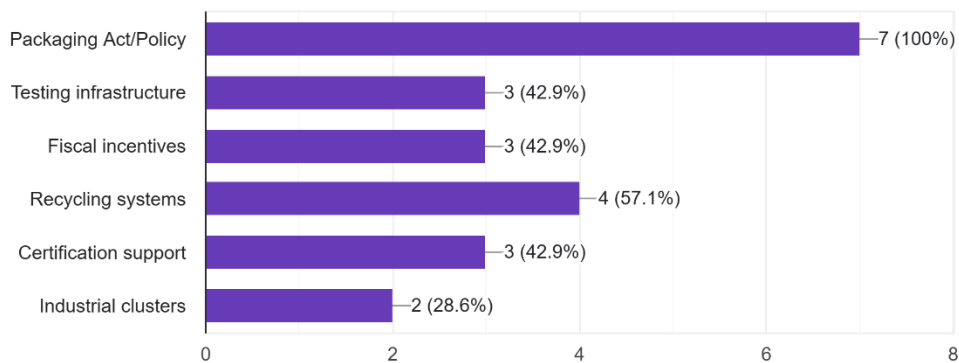
18. Are there significant skills or training gaps in packaging?



Training Gaps

- Waste management, PPWR regulations
- Limited expertise in circular economy packaging technology; intelligent packaging and shelf life extension technologies; as well as compliance to international standards
- Regulatory standards and compliance are nearly not existing or not in action..
- Limited technology
- Structured packaging training in Uganda is recent and limited

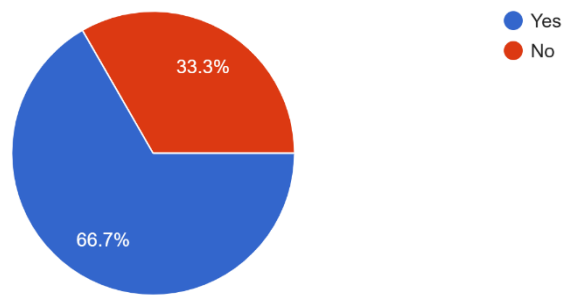
19. Which areas require urgent policy reform? (Select up to three)



20. Please provide any additional recommendations for strengthening Uganda's packaging value chain

- Packaging standards
- Government involvement
- create centers of excellence in packaging; rely on local raw materials for packaging, digitalize the packaging supply chain; harmonize standards; and strengthen quality infrastructure (testing, certification, inspection etc.)
- Government should reduce on taxes imposed on paper given the fluting local suppliers are not capable to sustain the players in the industry

Do you have the capacity to purchase packaging materials from outside Uganda



Please specify the type of Packaging Materials and Country

- East Africa
- India
- China
- Europe,
- Singapore
- Kenya